

Evaluation of Carbon Sink and Sequestration Potential of Trees within Hall-8

Field-based biomass estimation and age-based growth projection

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228

Trees surveyed

19

Species

3.7336 Mt

Current carbon stock CO₂e

1.318 Mt

CO₂e potential / year

Carbon Stock vs. Sequestration Potential

Two related but distinct measures used in the Hall-8 assessment

CARBON STOCK

The amount of carbon already stored in the biomass of the trees at a given point in time.

Think: “How much carbon is stored in the trees now?”

Hall-8: 3.7336 Mt CO₂e

SEQUESTRATION POTENTIAL

The estimated additional carbon that can be stored through future tree growth over a specified period.

Think: “How much additional carbon can the trees store as they grow?”

**Hall-8: 1.318 Mt
CO₂e/year**

CARBON STOCK = ALREADY STORED | SEQUESTRATION = ADDITIONAL STORAGE THROUGH GROWTH

1. Study Objectives

- Collect the primary data of tree and prepare an inventory of trees within Hall-8.
- Estimate above-ground biomass (AGB) and below-ground biomass (BGB).
- Estimate current carbon stock and convert it to CO₂-equivalent.
- Project subsequent-year biomass using an age–circumference growth relationship.
- Estimate annual carbon sequestration potential from projected biomass change.
- Identify species contributing most strongly to the Hall-8 carbon sink.

2. Field Inventory & Dataset

Primary field survey conducted within Hall-8

Specie Name	BW Block A-B	BW Block B-C	BW Block C-D	BW Block D-Mess	BW Block E-Mess	BW Block F - E	BW Block H - G	BW Hall office - H	Mess front	Outer road	Grand Total
Ailanthus excelsa	2	1		1						2	6
Artocarpus heterophyllus					1						1
Azadirachta indica		2	2	6	4	4	1		2	11	32
Bambusa sp.									2		2
Borassus flabellifer	1	2	2							1	6
Cassia fistula			1								1
Dalbergia sissoo	1	1	2	2						2	8
Delonix regia										1	1
Ficus religiosa										3	3
Mangifera indica	2	1	3	1	3	1	2	1		7	21
Manilkara zapota										1	1
Morus alba	3				1	1			1	1	7
Neolamarckia cadamba					1			1			2
Polyalthia longifolia										102	102
Psidium guajava	2	2	1	1		2	1		2		11
Syzygium cumini	1	1		2	3	1	1		1	4	14
Syzygium genus	1										1
Tectona grandis	1		3		1		2			1	8
Trapa natans									1		1
Grand Total	14	10	14	13	14	9	7	2	9	136	228

228

Trees

19

Species

1.37 m

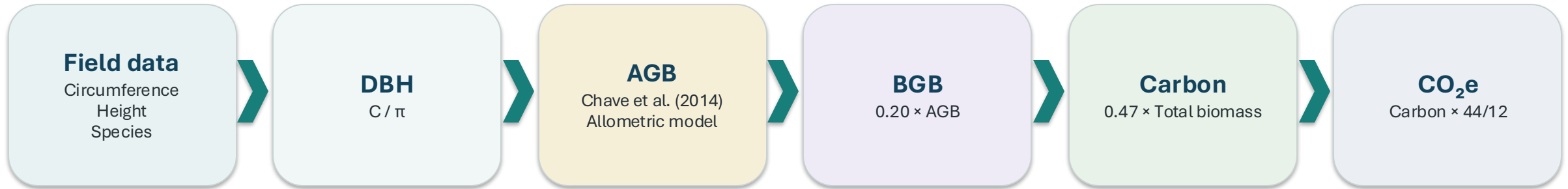
Measurement height

≥ 8 cm

Circumference included

3. Biomass & Carbon Estimation

Biomass-based carbon stock framework



$$\text{Total } AGB_{stems} = \sum_{i=1}^{228} (\rho_i \cdot D_i^2 \cdot H_i)^{0.976}$$

Where,

$\text{Total } AGB_{stems}$ = Total above-ground biomass of stems (kg)

ρ_i = Wood density of species i (kg m^{-3})

D_i = Diameter at breast height (DBH) of tree i (m)

H_i = Total height of tree i (m)

i = Number of trees ($i = 1$ to 228)

4. Age-Based Growth Projection

Estimating subsequent-year biomass where annual remeasurement was unavailable

- A Chapman–Richards growth function was fitted to available age–circumference observations.
- The fitted relationship was used to estimate age for trees without direct age observations.
- The derivative of the growth function provided an annual circumference growth increment.
- Projected circumference at $t+1$ was converted to DBH and then to biomass.
- Projected biomass was used to estimate subsequent-year carbon stock and CO_2 sequestration potential.

Chapman–Richards

$$C = K \cdot \left(1 - e^{-rA}\right)^{1/m}$$

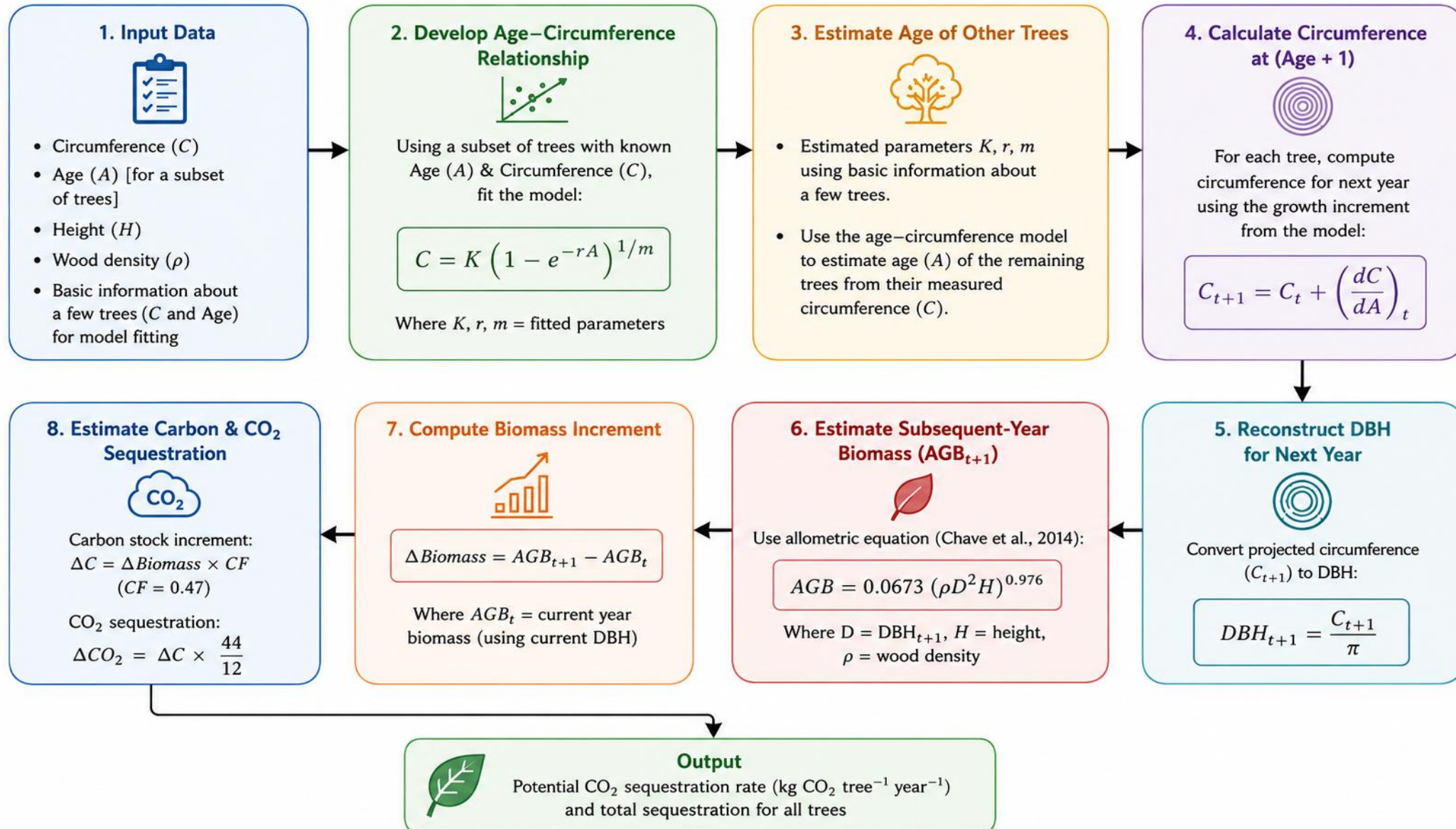
C = circumference

A = age

K, r, m = fitted parameters

$$C_{t+1} = C_t + \left(\frac{dC}{dA}\right)_t$$

5. Estimation of Sequestration Potential



6. Overall Carbon Stock & Sequestration Results

2.166 Mt

Total dry biomass

1.018.26 Mt C

Current carbon stock

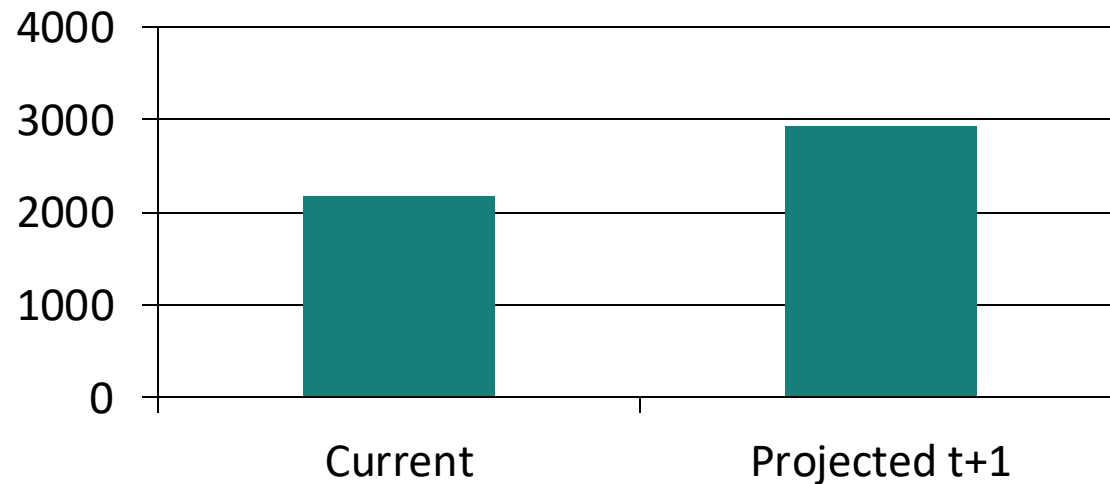
3.733 Mt

Stored CO₂-equivalent

1.318 Mt

CO₂ potential / year

Total biomass (kg)

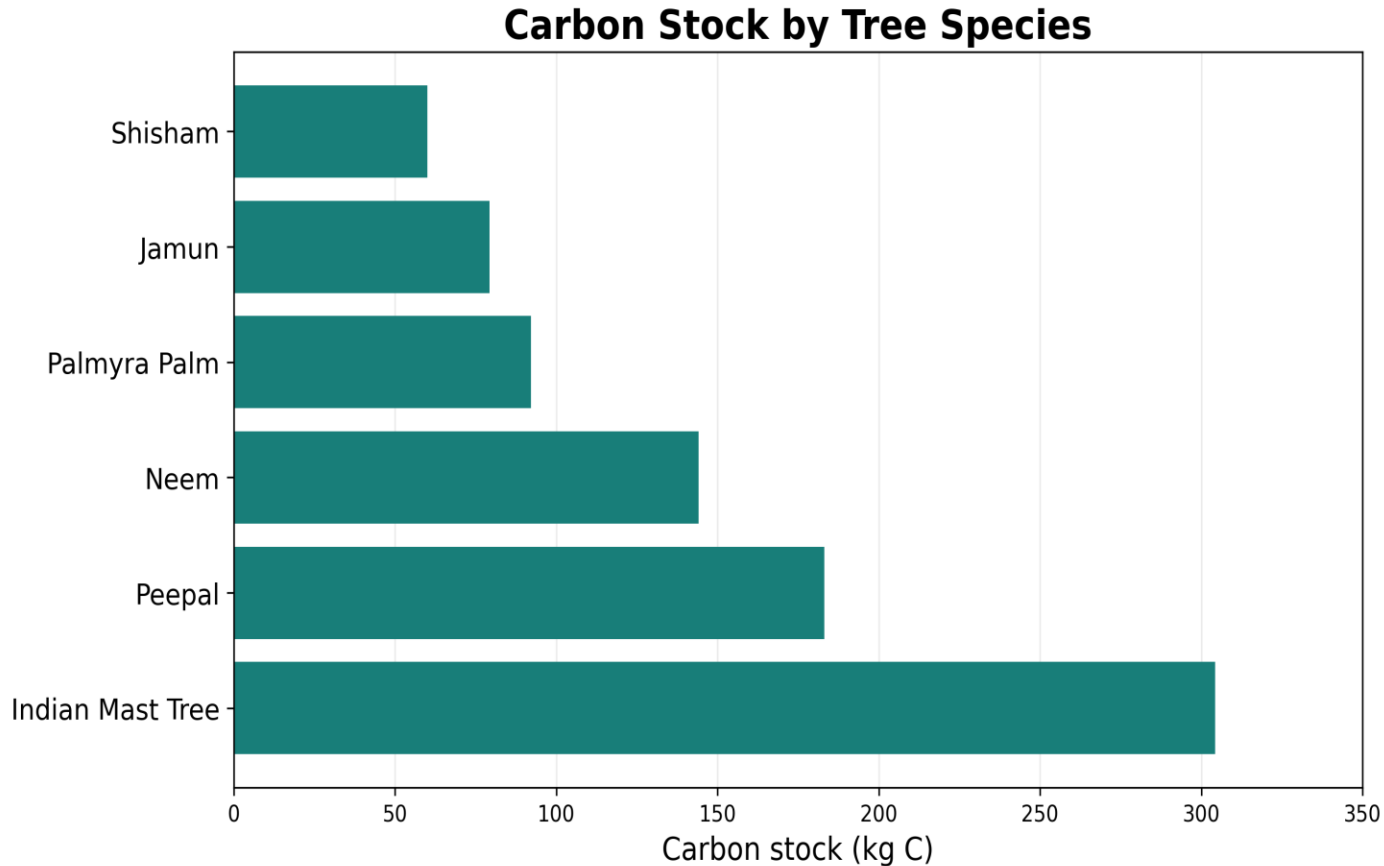


Key result

**Projected biomass increase:
765.07 kg over one year**

The sequestration figure is a model-based potential, not a directly observed annual change.

7. Which Species Contribute Most?



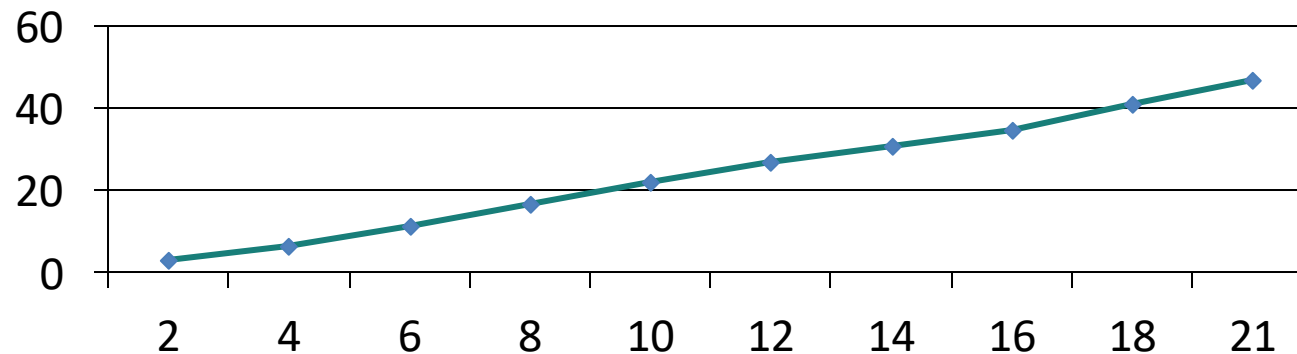
- Indian Mast Tree : 102 trees; 304.31 kg C — largest total contribution.
- Peepal : only 4 trees, but 183.20 kg C — high biomass per tree.
- Neem : 32 trees; 144.22 kg C.
- Indian Blackberry : 14 trees; 79.39 kg C.
- Tree abundance alone does not determine carbon storage; tree size and biomass are critical.

1 Metric tonne (Mt) = 1000 kg

8. DBH Growth with Time

- The theoretical growth relationship from the Chapman–Richards function was compared with observed Hall-8 age–DBH data.
- The theoretical curve represents a saturating forest-growth pattern, while the Hall-8 observations provide the empirical benchmark.
- The comparison supports the projection framework, but does not replace future field remeasurement.

Observed Hall-8 age–DBH



Interpretation

Growth tends to slow as tree size/age increases, motivating a saturating rather than purely linear growth representation.

9. Limitations & Interpretation

- Carbon stock depends on the accuracy of DBH and height measurements.
- Results are sensitive to the selected allometric equation and species wood-density values.
- Below-ground biomass was estimated using a generalized 20% BGB:AGB assumption rather than direct root measurement.
- A carbon fraction of 0.47 was applied to estimated dry biomass.
- Trees with circumference below 8 cm were excluded.
- The annual sequestration figure is model-based; actual annual sequestration should be validated through repeat measurements.

10. Conclusion

- Hall-8 contains 228 surveyed trees across 19 species, representing a measurable tree-based carbon sink.
- Estimated current carbon stock is 1.01826 Mt C ~ (3.733 Mt CO₂e).
- Model-based annual sequestration potential is 0.359 Mt C yr⁻¹, equivalent to 1.318 Mt CO₂ yr⁻¹.
- Indian Mast Tree is the largest contributor by total carbon stock, while *Ficus religiosa* illustrates the importance of individual tree biomass.
- The study establishes a baseline; repeat measurement of the same trees after one or more years is the preferred way to validate actual annual sequestration.

Thank You